Weekly Progress Report - Sprint 3

Group Name	MediVise
Week Number	9
Scrum Master	Sinem ÜNSAL
Date Submitted	15.03.2026
Domains Implemented So Far	17/20
Github URL	https://github.com/MhmtEmrKlnc/SENG430-project
Sprint	Sprint 4

1. Burndown Chart

2. Completed This Week

Story Id	Title	Story Points
US-020	Confusion matrix with clinical explanation	5
US-021	Display performance metrix cards	5
US-022	Global feature importance display	8
US-023	Local explanation for individual patient prediction	8

3. In Progress

Story Id	Title	Completion
US-024	Subgroup fairness comparison table	~90%
US-025	EU AI Act comliance checklist	~95%

4. Blocked / At Risk

- **Docker Orchestration Stability:** The containerization setup (docker-compose) is implemented but exhibits stability issues across different local environments. Thorough verification and debugging of Docker configurations are required to ensure the application reliably spins up for seamless evaluation and deployment.

5. Key Decisions Made

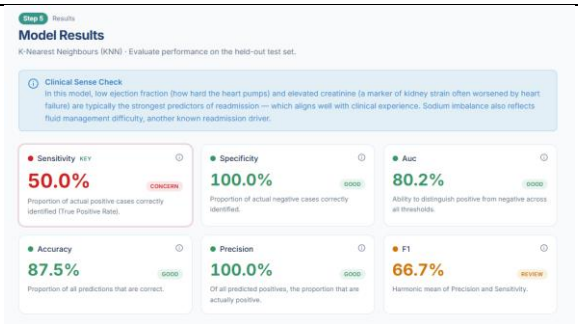
- **Detailed PDF Reporting Architecture:** Decided to decouple basic tracking from detailed reporting by integrating ReportLab's HorizontalBarChart. This allows detailed PDF exports to feature dynamic AI feature importance graphics, mapped with human-readable clinical labels.
- **Clinical Sense-Check Verification:** Decided to enforce clinical terminology across the UI and documentation (Domain_Clinical_Review.md), ensuring that AI explanations always prioritize medically relevant justifications (e.g., mapping raw database column age to "Patient Age").

6. Test Results

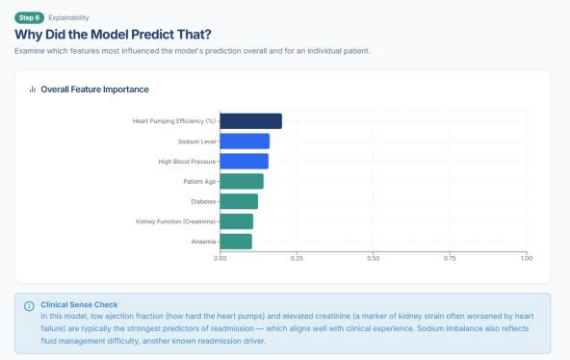
Title	Acceptence Criteria Status
US-020 Confusion matrix with clinical explanation	5
US-021 Display performance metric cards	5
US-022 Global feature importance display	8

US-023 Local explanation for individual patient prediction	8

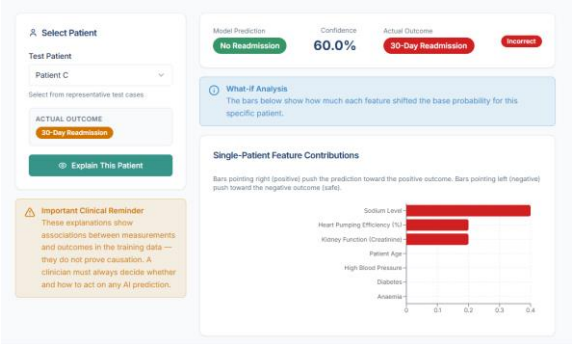
US-020



US-021



US-022



US-023

7. Metrics – Sprint 4

Metric	What to Measure	Target / Threshold
Usability Task Completion	Tasks T1–T7 completed by non-CS participant	≥ 5 of 7 tasks completed independently within time limits
SUS Score	System Usability Scale (10-question form)	≥ 68 (industry average)
Lighthouse Performance	Lighthouse audit on production build	≥ 80
Lighthouse Accessibility	Lighthouse accessibility audit	≥ 85
Docker Startup	docker-compose up timing	App fully loaded within 30 seconds
End-to-End Regression	Full Steps 1–7 with 3 different CSV files	Zero crashes across all 3 runs
Code Documentation	Functions with JSDoc / docstrings ÷ total functions	≥ 80% of functions documented

Full Domain Coverage	Test all 20 domains end-to-end (Steps 1–7)	All 20 domains complete the full pipeline without errors

8. Next Week Plan

- US-024 Finalize Subgroup fairness comparison table (Currently ~90% complete)
- US-025 Finalize EU AI Act compliance checklist (Currently ~95% complete)
- Task: Conduct end-to-end Usability Testing
- Task: Lighthouse Report Audit (Performance & Accessibility scores)
- Task: Resolve Docker configuration issues & verify final deployment

9. Retrospective Notes

Keep	Improve	Try
The team successfully expanded system support to all 20 domains while maintaining robust explainability logic. The integration of advanced visual graphics into the PDF exports was a major workflow success.	Deployment environment stability (Docker) needs to be standardized, so local testing and final evaluations are unaffected by system-specific inconsistencies.	In the upcoming final sprint, perform structured UI audits (Lighthouse) and extensive usability testing to ensure the project meets polished, production-ready quality standards.